

SYSTEM OVERVIEW

ASTRELIS NAV (SWARM PCB) is the electronics stack for an autonomous underwater vehicle (AUV) intended to operate as part of a swarm. It senses its own orientation, motion, and depth (Sensor board), processes this data and runs navigation/control logic (Control board), communicates with other units or a base station acoustically via an underwater piezo transducer (TX/RX board), drives its motors/servos/lights for propulsion and steering (Output board), and is powered from an onboard battery with regulated multi-rail power distribution (Power board) – all built as a compact stacked-board assembly suited to a small underwater housing.

Splitting SWARM into five functional boards (Control, Power, Sensor, TX/RX, Output) stacked via headers – rather than one monolithic board or fully separate, individually-housed boards – balances modularity with integration:

Noise isolation – Power (switching regulators) and TX/RX (sensitive RX analog, H-bridge) are separated from Sensor/Control, with isolated GND domains joined only at the stack connectors, limiting noise coupling.

Compact footprint – Stacking keeps the assembly small and tube-friendly, ideal for a compact vehicle.

Simple interconnect – Shared 47-pin headers pass signals board-to-board, avoiding custom wiring harnesses.

Modularity – Any board can be revised, repaired, or swapped independently without redesigning the whole system.

Easier fab/test – Each board can be built and tested on its own before final stack assembly.

CONTROL

CONTROL – System brain

STM32F446RET6 + Raspberry Pi Pico 2 run main flight logic and higher-level processing. Includes USB-C input (with ESD protection), microSD logging, SWD programming header, RGB/status LEDs, 16MHz/32.768kHz crystals, and a boot-mode jumper for the Pico.

POWER

POWER – Energy management

Takes battery input (XT30, 4S2P pack) with surge/reverse protection, regulates to 3.3V, 5V, and 12V via buck converters, plus a boosted VPiezo rail (TPS55340) for the transducer. Includes voltage-divider power monitoring, status LEDs, and supplies power-good signals to the rest of the stack.

SENSOR

SENSOR – Motion & environmental sensing

I2C cluster: two IMUs (ASM330LHHXTR, LSM6DS032TR), a magnetometer(MMC5603N)), and a depth/pressure sensor (MS583730BA01–50). Shares one I2C bus with interrupt lines back to Control, which it feeds with orientation/environmental data.

TX_RX

TX/RX – Acoustic communication

Two half-bridge MOSFET drivers (UCC27211) form an H-bridge driving the piezo transducer to transmit, with a precharge sequence at power-up. On receive, a preamp/instrumentation amp (INA823) and bandpass filter (20-30kHz, for FSK) feed a comparator; a T/R switch (ADG7421) isolates TX/RX paths, and an INA240 senses H-bridge current.

OUTPUT

OUTPUT – Actuation

Breaks out GPIO to ESC, servo, and light headers (5V rail, per-channel pull resistors), routing Control's commands to motors/actuators to actually move the vehicle.

HEADER_PINS – Test/Probe Access

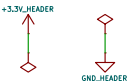
Breaks out stacking connector signals (J25-J28) to accessible pins for bench testing, mirroring each board's header pinout (3.3V, GND, GPIO, comms lines) without altering the signal path – lets you probe inter-board signals without disassembling the stack.

BOARD BREAKOUTS for testing

J25	J26	J27	J28
1 +3.3V_HEADER	1 3.3V	1 GPIO_0	1 GPIO_0
2 GPIO_28	2 GND_HEADER	2 GPIO_6	2 GPIO_6
3 GPIO_26	3 GPIO_27	3 GPIO_8	3 GPIO_8
4 PG_12V	4 PB14	4 GPIO_10	4 GPIO_10
5 PB15	5 PC6	5 GPIO_12	5 GPIO_12
6 PC7	6 PC8	6 GPIO_14	6 GPIO_14
7 PC9	7 PA9	7 LLA	7 LLA
8 PG_5V	8 LLA	8 PG_VPiezo	8 PG_VPiezo
9 PC10	9 PC11	9 ACC1_INT2	9 ACC1_INT2
10 GPIO_21	10 PB5	10 ADC_CURRENT	10 ADC_CURRENT
11 PG_3.3V	11 SDA	11 RX_FAULT	11 RX_FAULT
12 SCL	12 HI_B	12 ADC1_IN0	12 ADC1_IN0
		13 BPF_OUT	13 BPF_OUT

GND DOMAIN ARCHITECTURE
Each subsystem has an isolated GND domain: GND_POWER, GND_CONTROL, GND_SENSOR, GND_TXRX, GND_HEADER. These domains are connected through the stacking connector pins. No single star point.

PWR FLAGS



HEADER PINS

stacking connections between boards

HEADER_PINS		
+3.3V_HEADER	3.3V	PIN_23
GND_HEADER	GND	PIN_24
GPIO_28	PIN_1	GPIO_0
GPIO_27	PIN_2	GPIO_6
GPIO_26	PIN_3	GPIO_7
GPIO_22	PIN_4	GPIO_8
PG_12V	PIN_5	GPIO_9
PB14	PIN_6	GPIO_10
PB15	PIN_7	GPIO_11
PC6	PIN_8	GPIO_12
PC7	PIN_9	GPIO_13
PC8	PIN_10	GPIO_14
PC9	PIN_11	GPIO_15
PA9	PIN_12	LLA
PG_5V	PIN_13	PG_VPiezo
LI_B	PIN_14	ACC1_INT1
PC10	PIN_15	ACC1_INT2
PC11	PIN_16	ACC2_INT1
GPIO_21	PIN_17	ACC2_INT2
PB5	PIN_18	RX_FAULT
PG_3.3V	PIN_19	RX_EN
SDA	PIN_20	ADC1_IN0
SCL	PIN_21	ADC1_IN1
HI_B	PIN_22	BPF_OUT

DUPLICATE FOOTPRINTS BETWEEN BOARDS

By SAMANTHA CHONG

ASTRELIS NAV

Sheet: /

File: SWARM_PCB_Rev_2.kicad_sch

Title: SWARM PCB

Size: A4

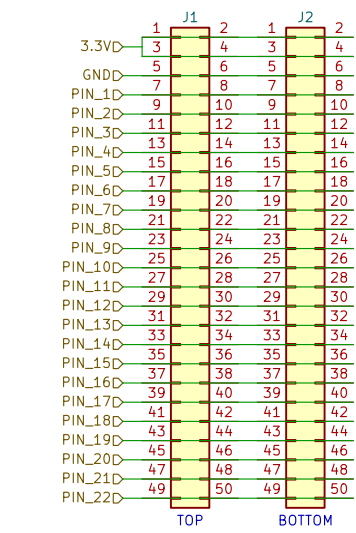
Date: 2026-06-19

KiCad E.D.A. 10.0.1

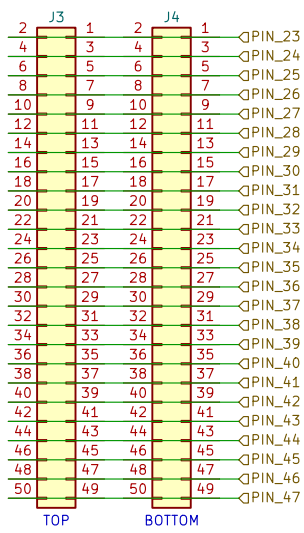
Rev: 2

Id: 1/12

LEFT SIDE HEADERS



RIGHT SIDE HEADERS



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ASTRELIS NAV

Sheet: /CONTROL/HEADER_PINS/

File: HEADER_PINS.kicad_sch

Title:

Size: A4

Date: 2026-06-19

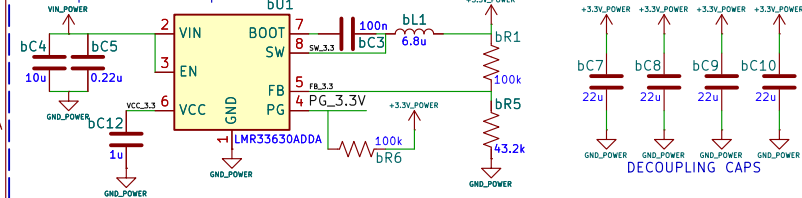
Rev:

KiCad E.D.A. 10.0.1

Id: 5/12

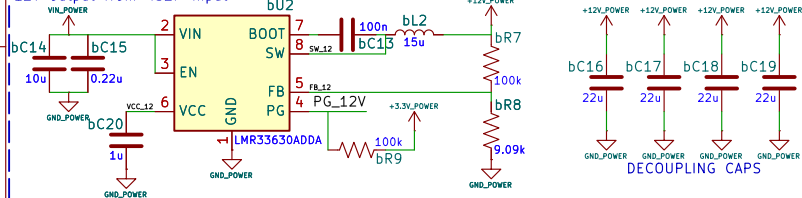
3.3V BUCK

3.3V output from 4S2P input



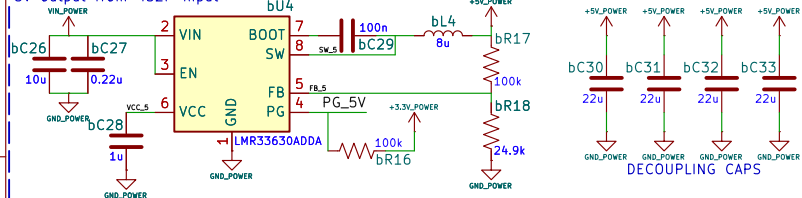
12V BUCK

12V output from 4S2P input



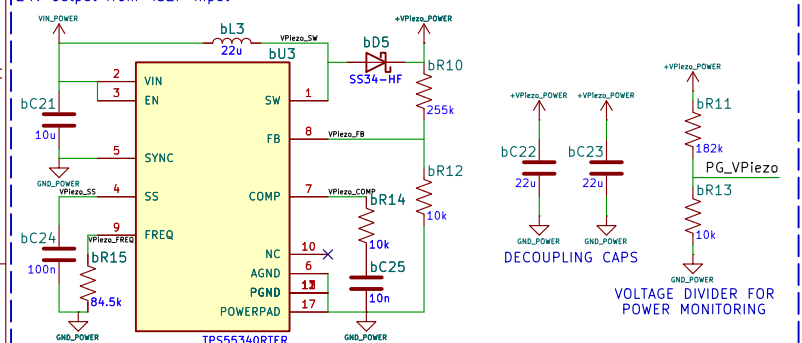
5V BUCK

5V output from 4S2P input



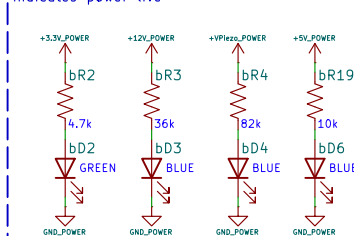
V_PIEZO BOOST

24V output from 4S2P input

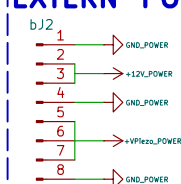


LEDs

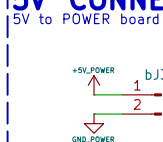
indicates power live



EXTERN POWER

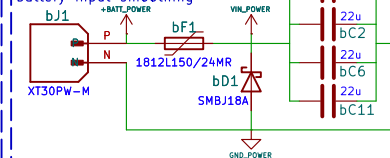


5V CONNECTOR

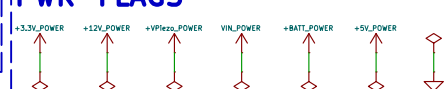


INPUT PROTECTION

battery input smoothing

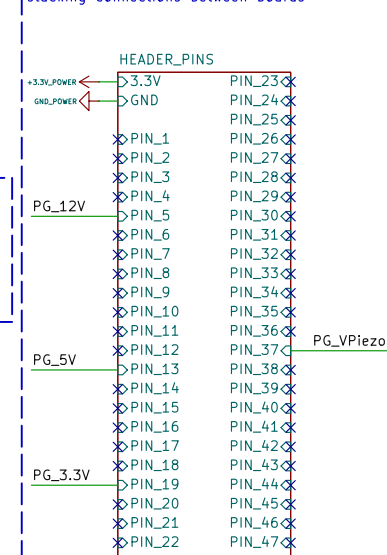


PWR FLAGS



HEADER PINS

stacking connections between boards



PG_3.3V → PB6 – digital, open-drain PG pin (pulled up via R6). LOW = 3.3V rail fault.

PG_5V → PA10 – digital, open-drain PG pin (pulled up via R16). LOW = 5V rail fault.

PG_12V → PB12 – digital, open-drain PG pin (pulled up via R9). LOW = 12V rail fault.

PG_VPiezo → PB1 (ADC1_IN9) – analog, R11/R13 voltage divider sampling VPiezo rail.

No digital flag – fault = ADC reading outside expected range.

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ASTRELIS NAV

Sheet: /POWER/

File: POWER.kicad_sch

Title: POWER BOARD

Size: A4

Date: 2026-06-19

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Rev: 2

Id: 3/12

LEFT SIDE HEADERS

	1	J5	2	1	J6	2
3.3V	3		4	3		4
GND	5		6	5		6
PIN_1	7		8	7		8
PIN_2	9		10	9		10
PIN_3	11		12	11		12
PIN_4	13		14	13		14
PIN_5	15		16	15		16
PIN_6	17		18	17		18
PIN_7	19		20	19		20
PIN_8	21		22	21		22
PIN_9	23		24	23		24
PIN_10	25		26	25		26
PIN_11	27		28	27		28
PIN_12	29		30	29		30
PIN_13	31		32	31		32
PIN_14	33		34	33		34
PIN_15	35		36	35		36
PIN_16	37		38	37		38
PIN_17	39		40	39		40
PIN_18	41		42	41		42
PIN_19	43		44	43		44
PIN_20	45		46	45		46
PIN_21	47		48	47		48
PIN_22	49		50	49		50
	TOP			BOTTOM		

RIGHT SIDE HEADERS

2	J7	1	2	J8	1	PIN_23
4		3	4		3	PIN_24
6		5	6		5	PIN_25
8		7	8		7	PIN_26
10		9	10		9	PIN_27
12		11	12		11	PIN_28
14		13	14		13	PIN_29
16		15	16		15	PIN_30
18		17	18		17	PIN_31
20		19	20		19	PIN_32
22		21	22		21	PIN_33
24		23	24		23	PIN_34
26		25	26		25	PIN_35
28		27	28		27	PIN_36
30		29	30		29	PIN_37
32		31	32		31	PIN_38
34		33	34		33	PIN_39
36		35	36		35	PIN_40
38		37	38		37	PIN_41
40		39	40		39	PIN_42
42		41	42		41	PIN_43
44		43	44		43	PIN_44
46		45	46		45	PIN_45
48		47	48		47	PIN_46
50		49	50		49	PIN_47
	TOP			BOTTOM		

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ASTRELIS NAV

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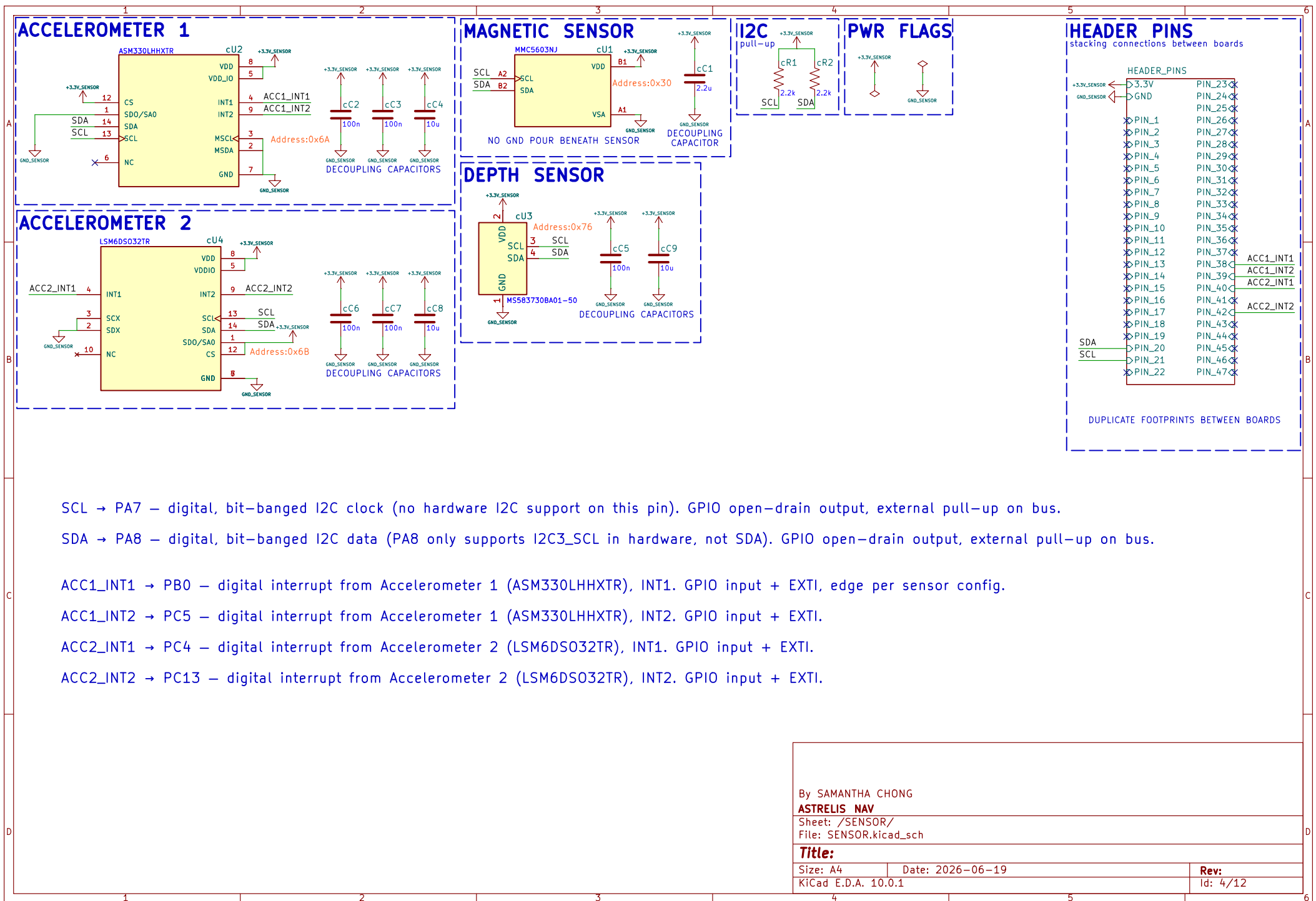
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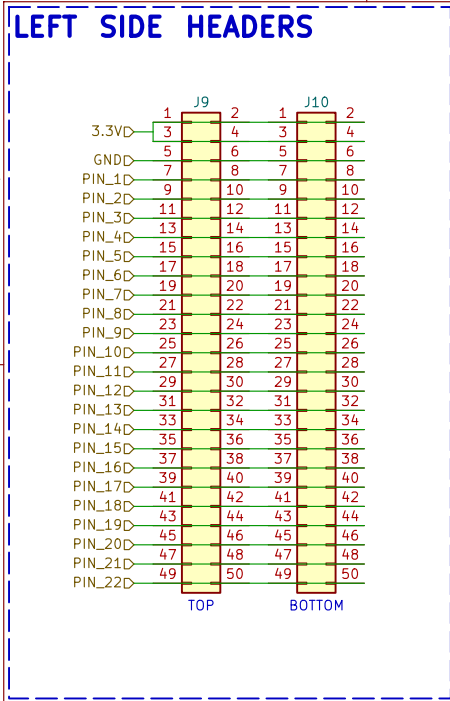
Rev:

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Id: 6/12

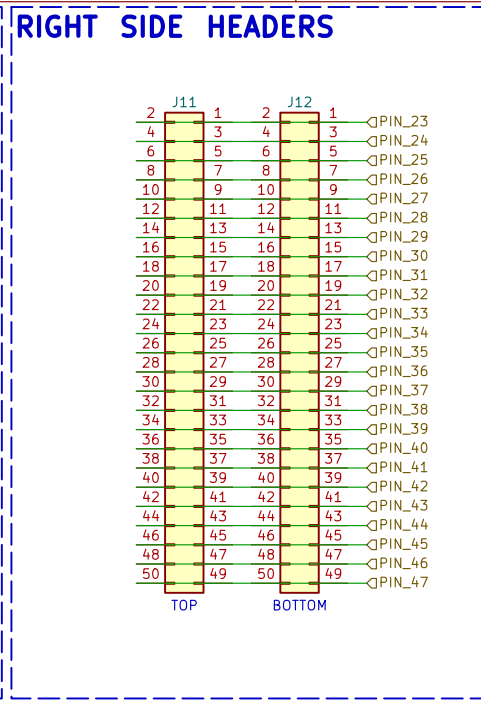


LEFT SIDE HEADERS					
	J9		J10		
3.3V	3	4	3	4	
GND	5	6	5	6	
PIN_1	7	8	7	8	
PIN_2	9	10	9	10	
PIN_3	11	12	11	12	
PIN_4	13	14	13	14	
PIN_5	15	16	15	16	
PIN_6	17	18	17	18	
PIN_7	19	20	19	20	
PIN_8	21	22	21	22	
PIN_9	23	24	23	24	
PIN_10	25	26	25	26	
PIN_11	27	28	27	28	
PIN_12	29	30	29	30	
PIN_13	31	32	31	32	
PIN_14	33	34	33	34	
PIN_15	35	36	35	36	
PIN_16	37	38	37	38	
PIN_17	39	40	39	40	
PIN_18	41	42	41	42	
PIN_19	43	44	43	44	
PIN_20	45	46	45	46	
PIN_21	47	48	47	48	
PIN_22	49	50	49	50	
	TOP		BOTTOM		

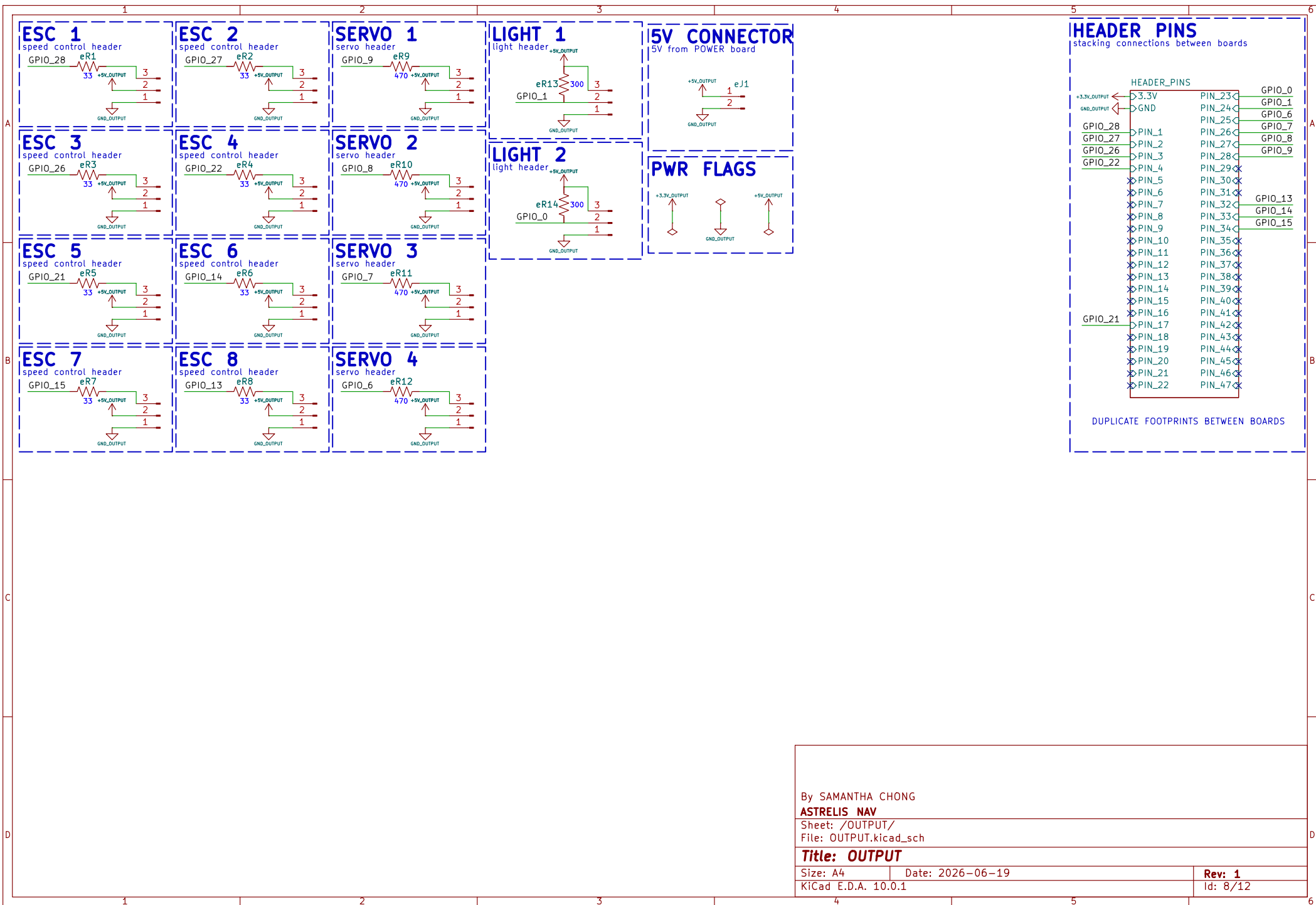


RIGHT SIDE HEADERS				
J11		J12		
2	1	2	1	
4	3	4	3	⇩PIN_23
6	5	6	5	⇩PIN_24
8	7	8	7	⇩PIN_25
10	9	10	9	⇩PIN_26
12	11	12	11	⇩PIN_27
14	13	14	13	⇩PIN_28
16	15	16	15	⇩PIN_29
18	17	18	17	⇩PIN_30
20	19	20	19	⇩PIN_31
22	21	22	21	⇩PIN_32
24	23	24	23	⇩PIN_33
26	25	26	25	⇩PIN_34
28	27	28	27	⇩PIN_35
30	29	30	29	⇩PIN_36
32	31	32	31	⇩PIN_37
34	33	34	33	⇩PIN_38
36	35	36	35	⇩PIN_39
38	37	38	37	⇩PIN_40
40	39	40	39	⇩PIN_41
42	41	42	41	⇩PIN_42
44	43	44	43	⇩PIN_43
46	45	46	45	⇩PIN_44
48	47	48	47	⇩PIN_45
50	49	50	49	⇩PIN_46

TOP BOTTOM



Id: 7/12



ESC 5

speed control header

GPIO_21

eR5

33

+5V_OUTPUT

3

2

1

GND_OUTPUT

ESC 6

speed control header

GPIO_14

eR6

33

+5V_OUTPUT

3

2

1

GND_OUTPUT

ESC 7

speed control header

GPIO_15

eR7

33

+5V_OUTPUT

3

2

1

GND_OUTPUT

ESC 8

speed control header

GPIO_13

eR8

33

+5V_OUTPUT

3

2

1

GND_OUTPUT

SERVO 1

servo header

GPIO_9

eR9

470

+5V_OUTPUT

3

2

1

GND_OUTPUT

SERVO 2

servo header

GPIO_8

eR10

470

+5V_OUTPUT

3

2

1

GND_OUTPUT

SERVO 3

servo header

GPIO_7

eR11

470

+5V_OUTPUT

3

2

1

GND_OUTPUT

SERVO 4

servo header

GPIO_6

eR12

470

+5V_OUTPUT

3

2

1

GND_OUTPUT

LIGHT 1

light header

GPIO_1

eR13

300

+5V_OUTPUT

3

2

1

GND_OUTPUT

LIGHT 2

light header

GPIO_0

eR14

300

+5V_OUTPUT

3

2

1

GND_OUTPUT

5V CONNECTOR

5V from POWER board

+5V_OUTPUT

eJ1

1

2

GND_OUTPUT

PWR FLAGS

+3.3V_OUTPUT

+5V_OUTPUT

GND_OUTPUT

HEADER PINS

stacking connections between boards

+3.3V_OUTPUT

GND_OUTPUT

GPIO_28

GPIO_27

GPIO_26

GPIO_22

GPIO_21

GPIO_15

PIN_23

PIN_24

PIN_25

PIN_26

PIN_27

PIN_28

PIN_29

PIN_30

PIN_31

PIN_32

PIN_33

PIN_34

PIN_35

PIN_36

PIN_37

PIN_38

PIN_39

PIN_40

PIN_41

PIN_42

PIN_43

PIN_44

PIN_45

PIN_46

PIN_47

GPIO_0

GPIO_1

GPIO_6

GPIO_7

GPIO_8

GPIO_9

GPIO_13

GPIO_14

GPIO_15

DUPLICATE FOOTPRINTS BETWEEN BOARDS

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ASTRELIS NAV

Sheet: /OUTPUT/

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Title: OUTPUT

Size: A4

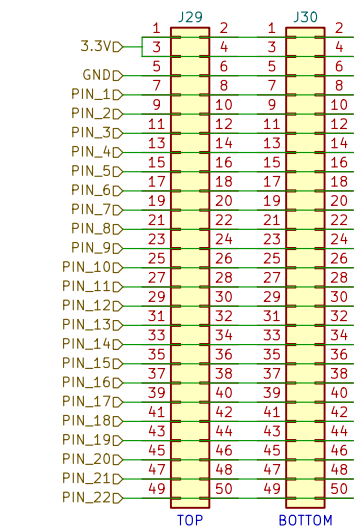
Date: 2026-06-19

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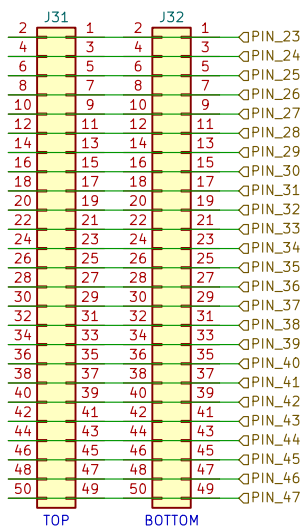
Rev: 1

Id: 8/12

LEFT SIDE HEADERS



RIGHT SIDE HEADERS



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ASTRELIS NAV

Sheet: /OUTPUT/HEADER_PINS/

File: HEADER_PINS.kicad_sch

Title:

Size: A4

Date: 2026-06-19

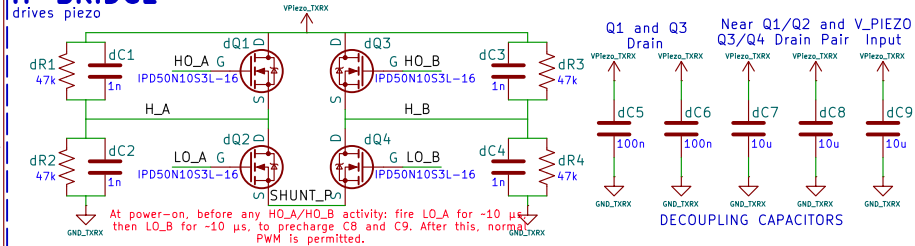
Rev:

KiCad E.D.A. 10.0.1

Id: 10/12

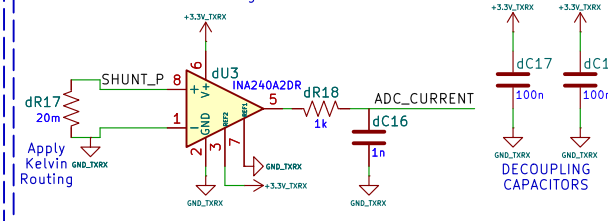
H-BRIDGE

drives piezo



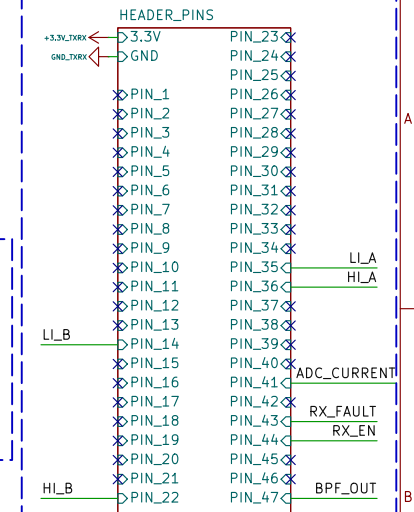
CURRENT SENSE

measures current from H-bridge

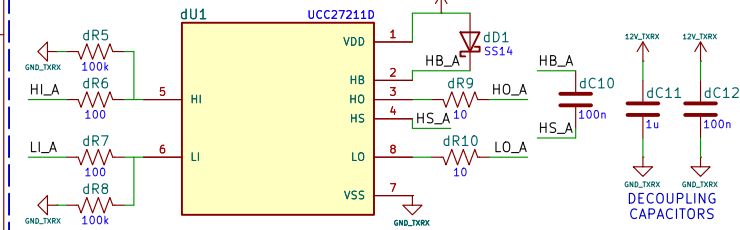


HEADER PINS

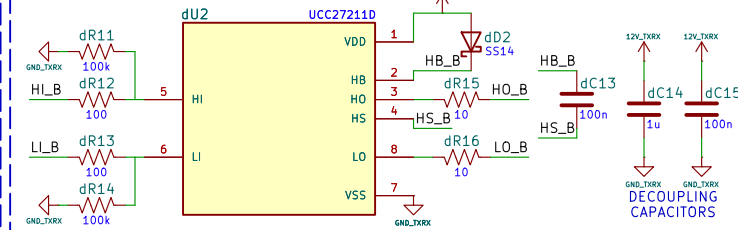
stacking connections between boards



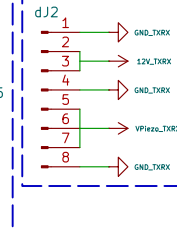
H-BRIDGE DRIVER A



H-BRIDGE DRIVER B

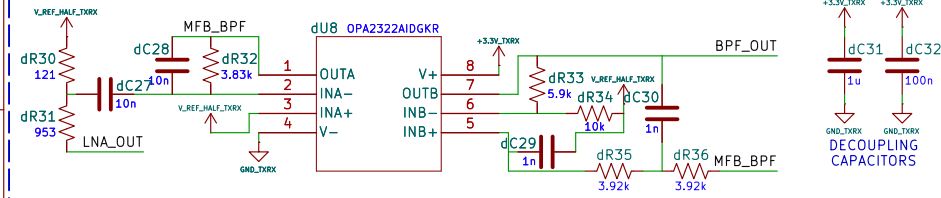


EXTERN



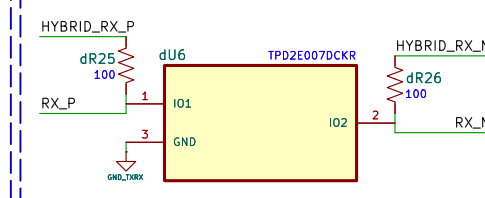
BAND PASS FILTER

20 kHz–30 kHz bandpass for FSK demodulation

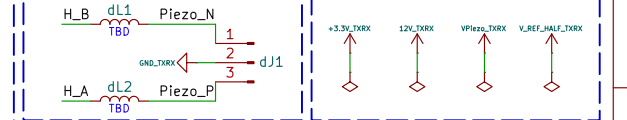


INPUT PROTECTION

ESD protection for differential RX input

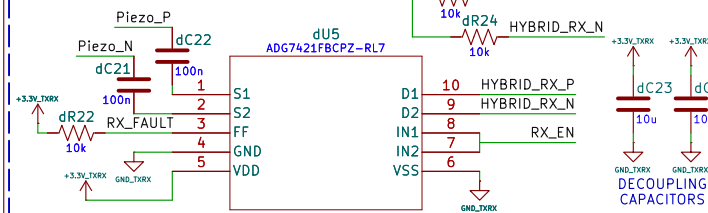


IMPEDANCE MATCH PWR FLAGS



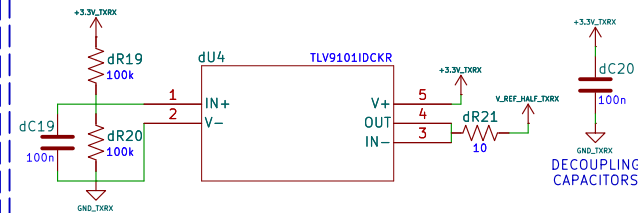
T/R SWITCH

TX/RX isolation switch



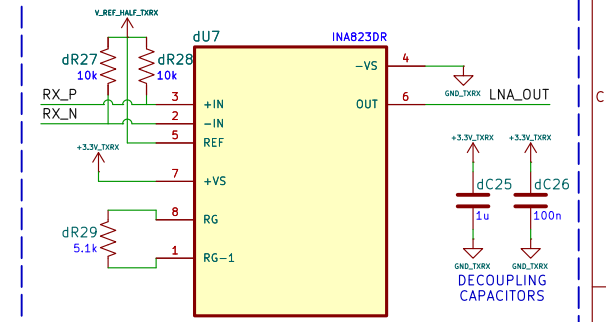
V_REF BUFFER

generates 1.65V output



RX PREAMP

fixed-gain instrumentation amplifier



RX_EN → PC3 – switch-enable output. HIGH = TX/RX switch enabled for receive mode.
RX_FAULT → PC0 – fault flag input from TX/RX isolation switch. LOW = switch fault.

ADC_CURRENT → PA3 (ADC1_IN3) – analog, INA240 current-sense output monitoring H-bridge current.
BPF_OUT → PA2 (ADC1_IN2) – analog, output of 20-30kHz bandpass filter (FSK demod stage).

LI_A → PA5 (TIM2_CH3) – digital PWM output, low-side gate drive to H-bridge driver A.
LI_B → PB0 (TIM2_CH1) – digital PWM output, low-side gate drive to H-bridge driver B.
HI_A → PA13 (TIM2_CH4) – digital PWM output, high-side gate drive to H-bridge driver A.
HI_B → PA6 (TIM2_CH2) – digital PWM output, high-side gate drive to H-bridge driver B.

By SAMANTHA CHONG

ASTRELIS NAV

Sheet: /TX_RX/

File: TX_RX.kicad_sch

Title: TX & RX

Size: A4 Date: 2026-06-19

KiCad E.D.A. 10.0.1

Rev: 2

Id: 9/12

LEFT SIDE HEADERS

	1	J17	2	1	J18	2
3.3V	3		4	3		4
GND	5		6	5		6
PIN_1	7		8	7		8
PIN_2	9		10	9		10
PIN_3	11		12	11		12
PIN_4	13		14	13		14
PIN_5	15		16	15		16
PIN_6	17		18	17		18
PIN_7	19		20	19		20
PIN_8	21		22	21		22
PIN_9	23		24	23		24
PIN_10	25		26	25		26
PIN_11	27		28	27		28
PIN_12	29		30	29		30
PIN_13	31		32	31		32
PIN_14	33		34	33		34
PIN_15	35		36	35		36
PIN_16	37		38	37		38
PIN_17	39		40	39		40
PIN_18	41		42	41		42
PIN_19	43		44	43		44
PIN_20	45		46	45		46
PIN_21	47		48	47		48
PIN_22	49		50	49		50
	TOP		BOTTOM			

RIGHT SIDE HEADERS

2	J19	1	2	J20	1	
4		3	4		3	PIN_23
6		5	6		5	PIN_24
8		7	8		7	PIN_25
10		9	10		9	PIN_26
12		11	12		11	PIN_27
14		13	14		13	PIN_28
16		15	16		15	PIN_29
18		17	18		17	PIN_30
20		19	20		19	PIN_31
22		21	22		21	PIN_32
24		23	24		23	PIN_33
26		25	26		25	PIN_34
28		27	28		27	PIN_35
30		29	30		29	PIN_36
32		31	32		31	PIN_37
34		33	34		33	PIN_38
36		35	36		35	PIN_39
38		37	38		37	PIN_40
40		39	40		39	PIN_41
42		41	42		41	PIN_42
44		43	44		43	PIN_43
46		45	46		45	PIN_44
48		47	48		47	PIN_45
50		49	50		49	PIN_46
	TOP		BOTTOM			PIN_47

By SAMANTHA CHONG

ASTRELIS NAV

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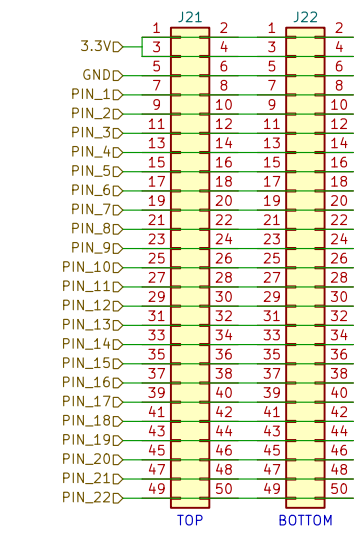
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Rev:

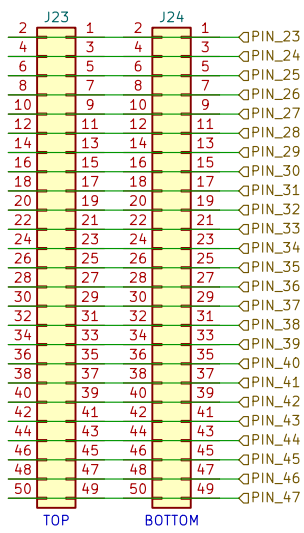
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LEFT SIDE HEADERS



RIGHT SIDE HEADERS



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ASTRELIS NAV

Sheet: /HEADER_PINS/

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Date: 2026-06-19

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